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PROFILE

MSc Data Science and BSc Mathematics graduate with a strong foundation in machine learning, statistics, data analysis, and predictive modelling. Experienced in analysing complex datasets, building machine learning solutions, developing dashboards, and delivering actionable insights through data-driven decision-making. Proficient in Python, SQL, Excel, Power BI, and modern machine learning techniques, with academic experience spanning NLP, computer vision, forecasting, and AI ethics. Strong analytical thinker with excellent problem-solving skills, attention to detail, and a keen interest in responsible AI, emerging technologies, and business-focused analytics.

EDUCATION

MSc. Data Science with Advanced Research (Distinction), University of Hertfordshire, Hatfield, UK (Jan 2023 – Mar 2025)

- Developed analytical, modelling, and data visualization skills through large-scale data projects and applied research.
- Thesis: Predicting Movie Ratings and Analyzing Feature Importance Using Tags and Genres
- Relevant Modules: Statistical Modelling, Python, SQL, Data Analysis, Data Visualization, Time Series Analysis, Machine Learning, AI Ethics.

BSc. Mathematics (Distinction), St. Berchmans College, Kerala, India (Jun 2019 – Sept 2022)

- Strong foundation in Probability, Statistics, Hypothesis Testing, Calculus, Linear Algebra, Differential Equations, and Economics.
- Developed rigorous quantitative reasoning and problem-solving skills applicable across analytics, finance, actuarial science, and risk management.

TECHNICAL SKILLS

- **Programming:** Python (Pandas, NumPy, Scikit-Learn), R, SQL, Excel (Pivot Tables, VBA, Power Query).
- **Reporting:** Microsoft Word, Powerpoint, LaTeX, Power BI, Tableau, Matplotlib, Seaborn.
- **Quantitative & Statistical Skills:** Statistical Modelling, Regression Analysis, Time Series Analysis, Predictive Analytics, Risk Analysis, Data Validation, Financial Reconciliation, Trend Analysis, Reporting, Process Automation.
- **Other Tools:** Microsoft Office suites, GitHub, Jupyter Notebook, Microsoft Azure.

PROJECTS

Predicting Movie Ratings and Analyzing Feature Importance Using Tags and Genres

- Developed deep machine models in Python to predict movie ratings using genres and user-generated tags from the MovieLens dataset.
- Performed data preprocessing, exploratory data analysis and feature engineering on large datasets.
- Applied feature importance techniques to identify factors most strongly influencing user ratings.
- Evaluated model performance using statistical metrics and visual analysis, providing insights into viewer preferences and recommendation systems.

Fine-Tuning DistilBERT for Toxicity Classification

- Fine-tuned a DistilBERT transformer model using TensorFlow(Python) to detect toxic and harmful online content.
- Implemented tokenization, text preprocessing, and model optimisation techniques for NLP classification.
- Evaluated model performance using ROC-AUC score and other classification metrics.
- The method showed a high recall for toxic comments while occasionally labeling non-toxic comments as toxic, prioritizing identifying all potential toxicity over preventing false positives.

Image Segmentation with Deep Learning

- Developed a convolutional neural network for image segmentation using a curated subset of the COCO 2017 dataset.
- Applied image preprocessing, augmentation, and training techniques to improve model performance.
- Evaluated segmentation quality using Intersection over Union (IoU) metrics.
- Visualized predicted segmentations to assess object boundary detection in complex scenes.

Financial Market Forecasting

- Developed forecasting models using historical market data to analyze share price movements.
- Conducted time-series analysis and data preprocessing to identify market trends and predictive factors.
- Assessed model performance using statistical error metrics against baseline forecasting methods and sensitivity analysis to evaluate forecasting reliability.
- Built Power BI dashboards to communicate investment insights and model results.

PROFESSIONAL EXPERIENCE

Night Auditor and Guest Service Assistant | Holiday Inn Express (IHG Brand), Watford, UK **(June 2023 – Present)**

- Reconcile daily financial transactions and operational records, ensuring reporting accuracy and data integrity.
- Investigate and resolve discrepancies across systems while maintaining strong financial controls.
- Produce audit reports and operational summaries within strict reporting deadlines.
- Analyze occupancy, revenue, and customer trends through Power BI dashboards to support business decision-making.
- Automated welcome letter generation for IHG members using Excel and Word, incorporating tier-level benefits and alphabetical ordering, replacing a manual system.
- Collaborated with finance, operations, reservations, and customers to resolve issues and maintain service quality.
- Handle confidential information while maintaining high standards of professionalism and accuracy.
- Work effectively in a fast-paced environment where accuracy and timely completion of tasks are critical.

Mathematics Tutor | Universal Tuition Centre, Kerala, India

(May 2021 - Jan 2023)

- Delivered mathematics and statistics tuition to students with diverse learning needs.
- Simplified complex quantitative concepts through structured problem-solving approaches.
- Developed strong communication, presentation, and mentoring skills.

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

- Institute and Faculty of Actuaries (IFoA) – Student Member
- CS1(Actuarial Statistics) sat in April 2026 – Results pending
- Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure AI Fundamentals

VOLUNTEERING & ACTIVITIES

- Active committee member of the Physics, Astronomy and Mathematics Society, Sim Racing Society, and Anime Society at the University of Hertfordshire.
- Organized events and collaborated with diverse teams to deliver successful student activities.
- Led the Rubik's Cube competition during college events, demonstrating leadership, organization, and teamwork skills.
- Participated in and won mathematics competitions throughout school and university.

ADDITIONAL INFORMATION

Languages - Fluent in *English*, Hindi, Malayalam.

Hobbies and Interests - Personal Finance | Financial Markets | Rubik's Cubes | Playing Keyboard | Football | Cricket | Motorsports | Sim Racing | Movies | Music